
Rachel Sunarko

Portfolio.

2025

sunarkorachel@gmail.com
085701933591
Data Science Undergraduate
Visual Storyteller | UI/UX Designer

I am second-year data science undergraduate with strong experience in visual storytelling and digital design. Passionate about creating interactive and visually appealing user interfaces with a strong sense of aesthetics and usability. Experienced in managing student-led projects, leading creative teams, and participating in national design competitions.

Highlights

- Developed the PackMe website as a frontend developer for a programming algorithm project, earning an A grade and presenting it to students from Universiti Malaya and Nanyang Polytechnic during the Student Inbound Program.
- Designed and implemented a responsive landing page for PackMe using HTML, CSS, and JavaScript, ensuring smooth frontend-backend integration.
- Led creative direction for BINARY 4.0, managing event branding across platforms including animated content, Instagram feeds, and visual identity.
- Participated in 10+ national infographic and visual storytelling competitions, applying data-driven design principles and earning multiple honorary placements.

Hard Skills:

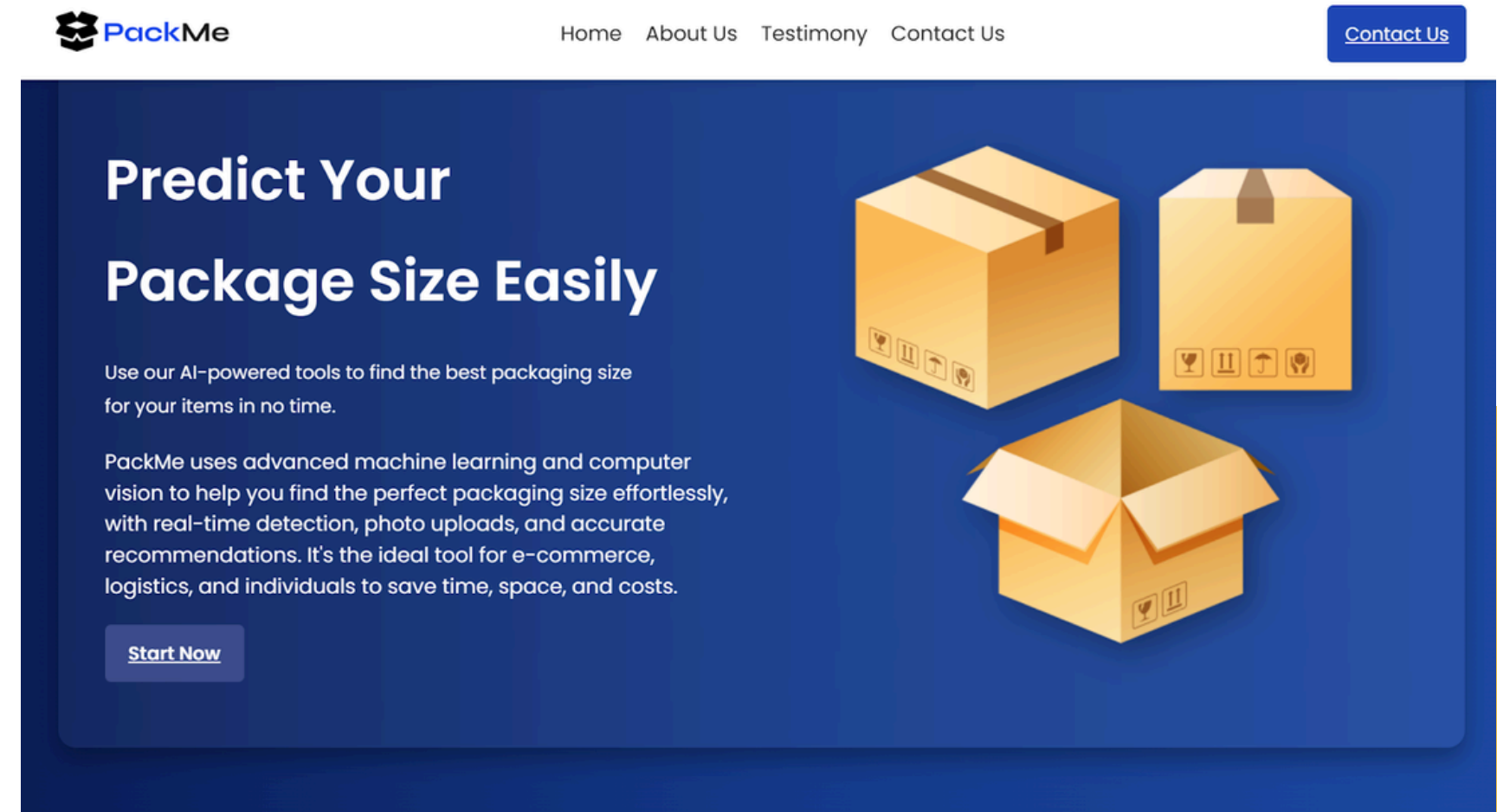
- Programming Languages (Python, C++, SQL, R, HTML, CSS, JavaScript)
- Data & Analysis Tools (Google Colaboratory, Jupyter Notebook, R Studio, Visual Studio Code, Tableau, Microsoft Excel)
- Version Control & Platforms (GitHub)
- Design & Multimedia Tools (Adobe Illustrator, Adobe Premiere Pro, Adobe After Effects, Adobe InDesign, Canva, Figma)
- Office Tools (Microsoft Word, Microsoft PowerPoint, Microsoft Excel)

Soft Skills:

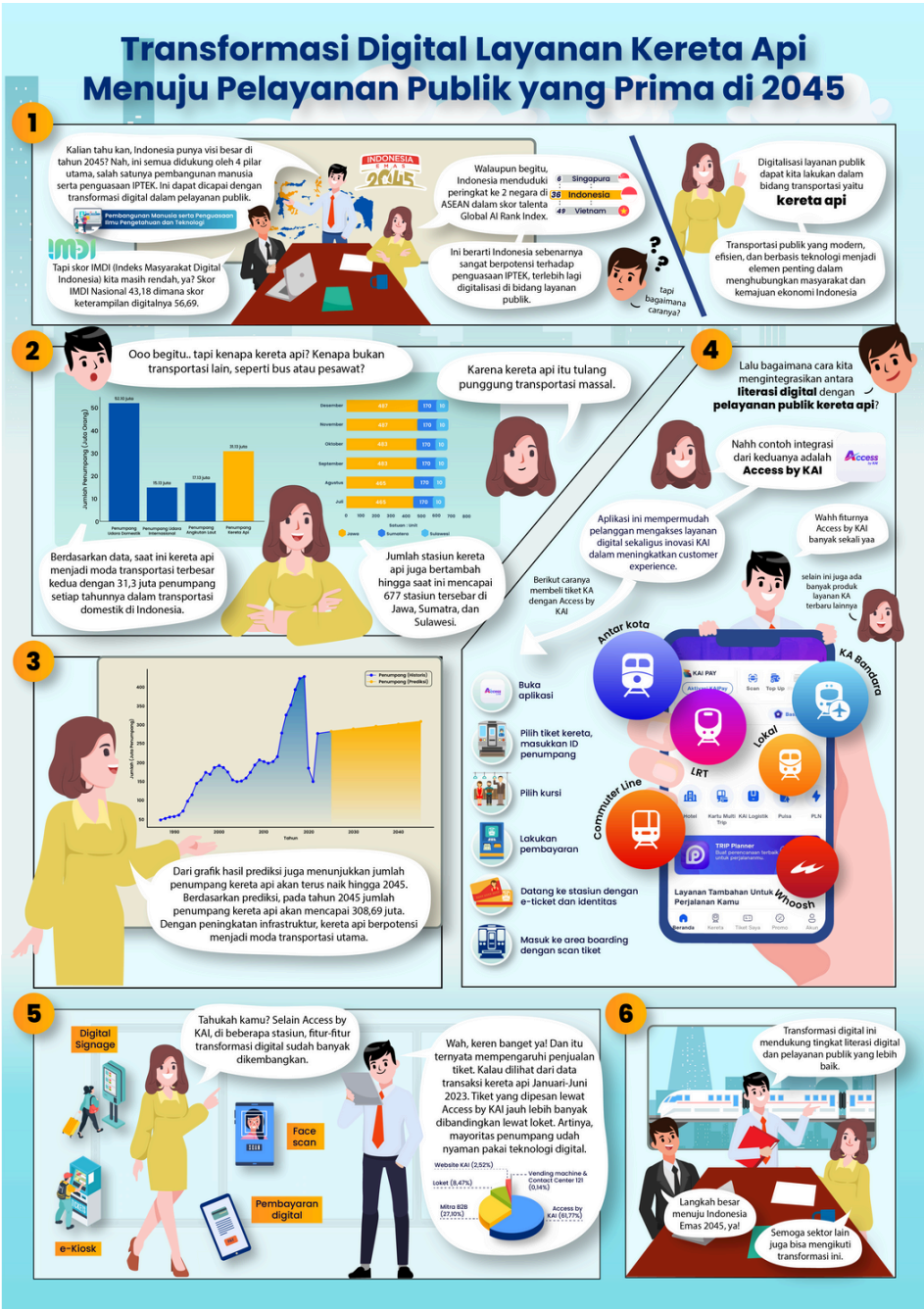
Teamwork & Leadership, Logical & Innovative Thinking, Leadership, Strong Work Ethic, Adaptability, Problem-Solving, Work Well Under Pressure

PackMe – Object Size Prediction and Packaging Recommendation Web-App

Designed and developed a responsive landing page for PackMe, a web application that predicts package sizes based on user input. The interface was built using HTML, CSS, and JavaScript, applying clean, modern UI principles to ensure visual appeal and ease of use. Emphasis was placed on intuitive layout design and smooth user interaction, while close collaboration with backend developers ensured seamless integration between the front-end interface and the underlying prediction logic.

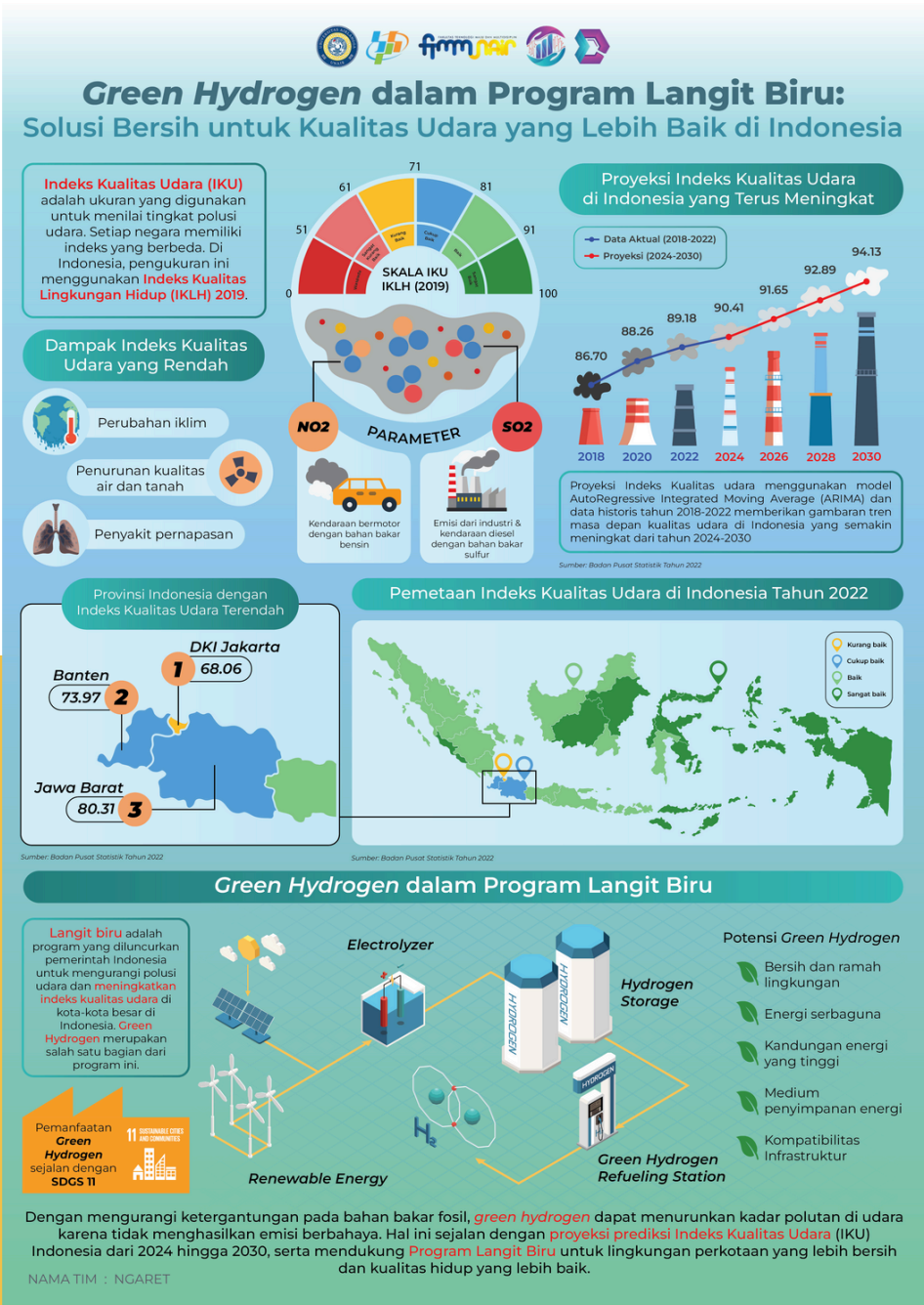


Infographic – Transformasi Digital Layanan Kereta Api Menuju Pelayanan Publik yang Prima di 2045



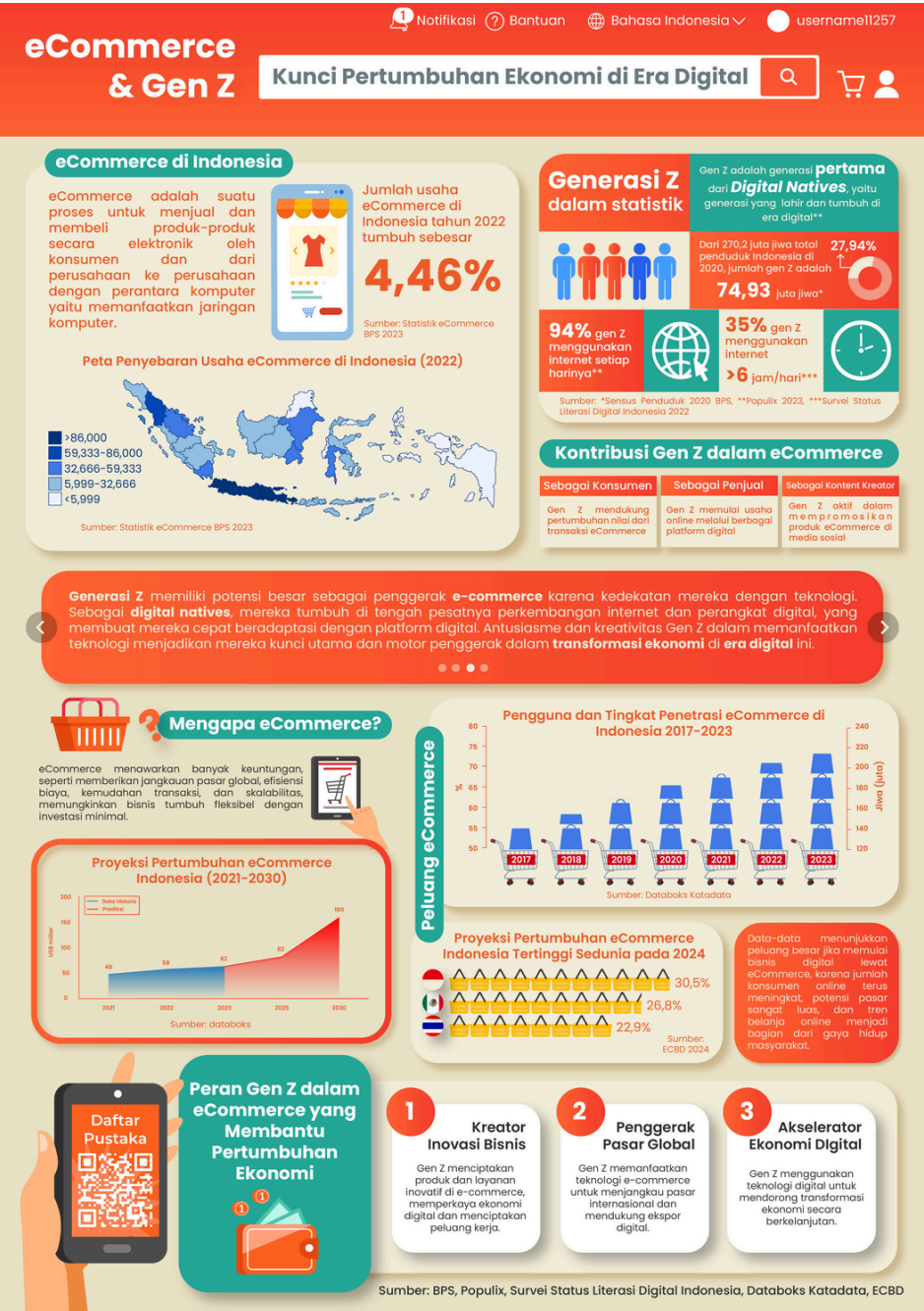
Created an infographic on Indonesia's railway digital transformation using data analysis and visual storytelling. Analyzed trends in passenger volume, station activity, and digital readiness, and applied exponential smoothing for future projections. Translated complex data into clear visuals aligned with the Indonesia Emas 2045 vision, enhancing public understanding through engaging, data-driven narratives.

Infographic – Green Hydrogen dalam Program Langit Biru: Solusi Bersih untuk Kualitas Udara yang Lebih Baik di Indonesia



Designed an infographic on air quality in Indonesia, applying time-series forecasting (ARIMA) to project future Air Quality Index (IKU) trends from 2024 to 2030. Mapped regional air quality data and analyzed pollution sources (NO₂, SO₂) to support the case for green hydrogen solutions. Effectively communicated complex environmental data into a clear, visually engaging narrative aligned with sustainable development goals (SDGs).

Infographic – eCommerce & Gen Z: Kunci Pertumbuhan Ekonomi di Era Digital



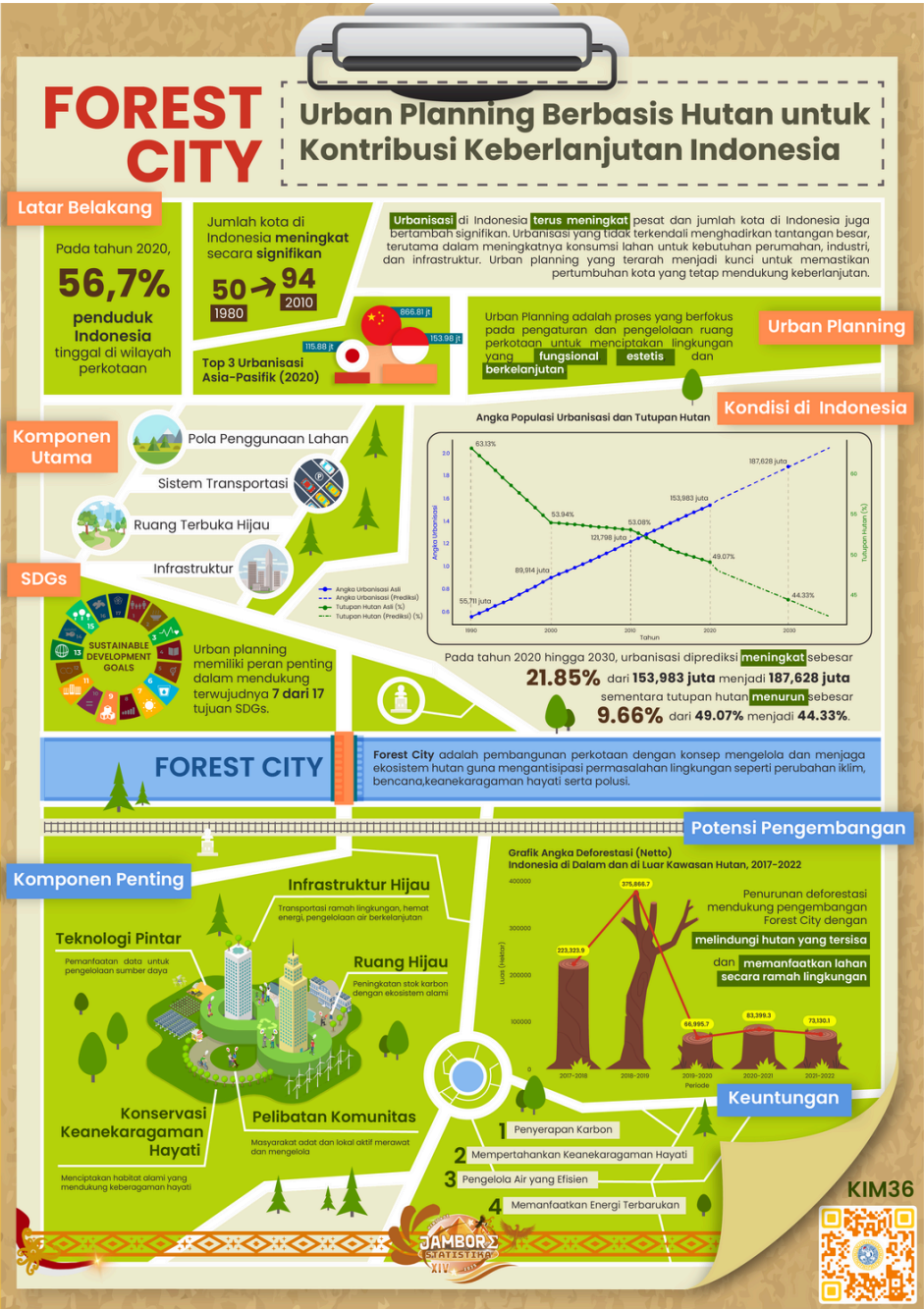
Created a data-driven infographic showcasing Gen Z’s pivotal role in Indonesia’s eCommerce growth using multi-source statistics (BPS, Populix, Katadata). Visualized temporal growth trends (2017–2023), regional eCommerce distribution, and projections to 2030. Combined demographic insights with digital behavior to narrate a compelling story of economic transformation through Gen Z participation.

Infographic – Buka Data, Bongkar Dusta: Lawan Isu Hoaks Pemilu dengan Data Journalism



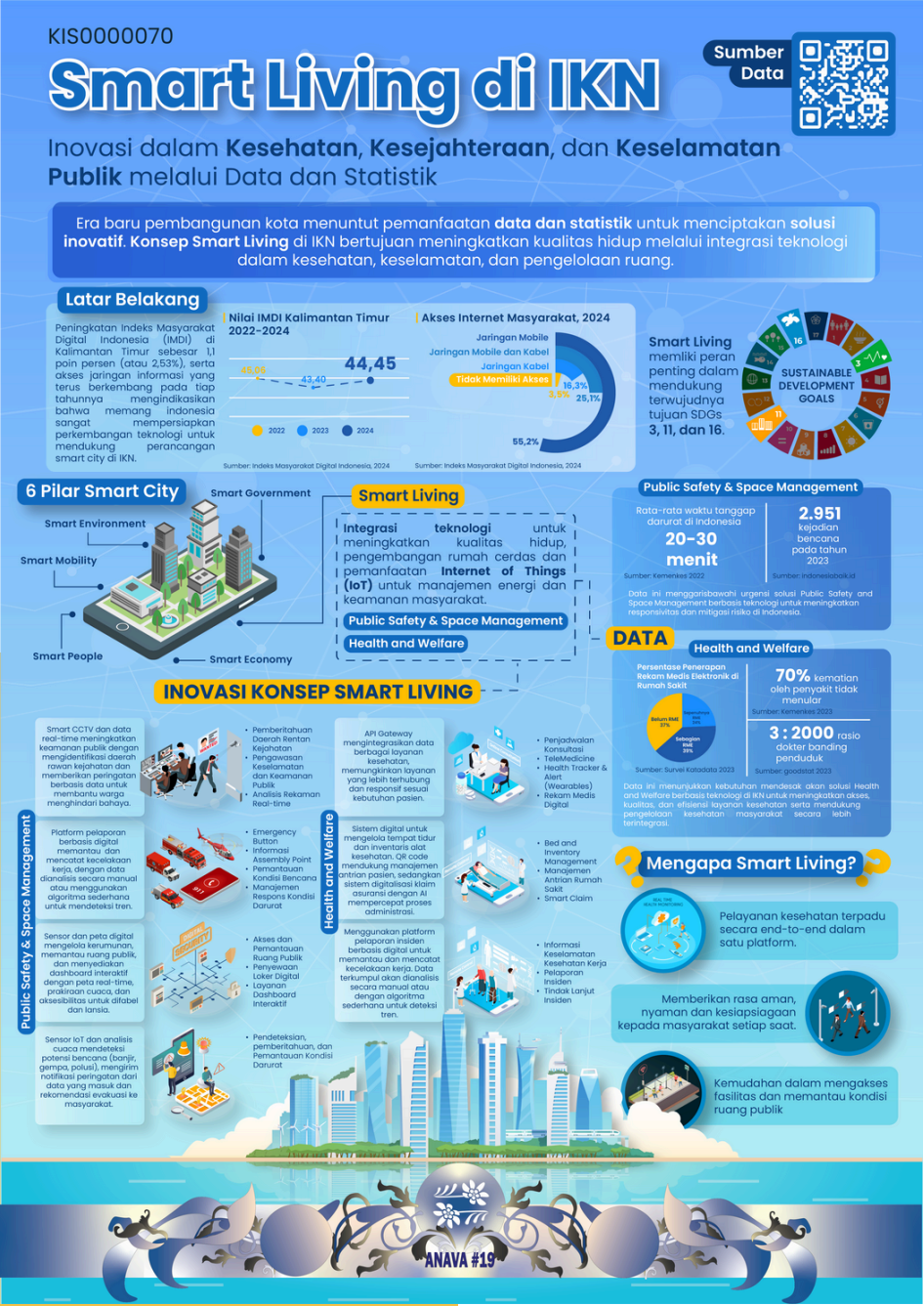
Created a compelling data journalism piece dissecting political hoaxes during the 2024 Indonesian election using clustering, word cloud analysis, and time series charts. Applied advanced data filtering and visualization techniques to expose misinformation patterns, highlight media platforms with high hoax incidence, and communicate verified facts in an accessible narrative format.

Infographic – Forest City: Urban Planning Berbasis Hutan untuk Kontribusi Keberlanjutan Indonesia



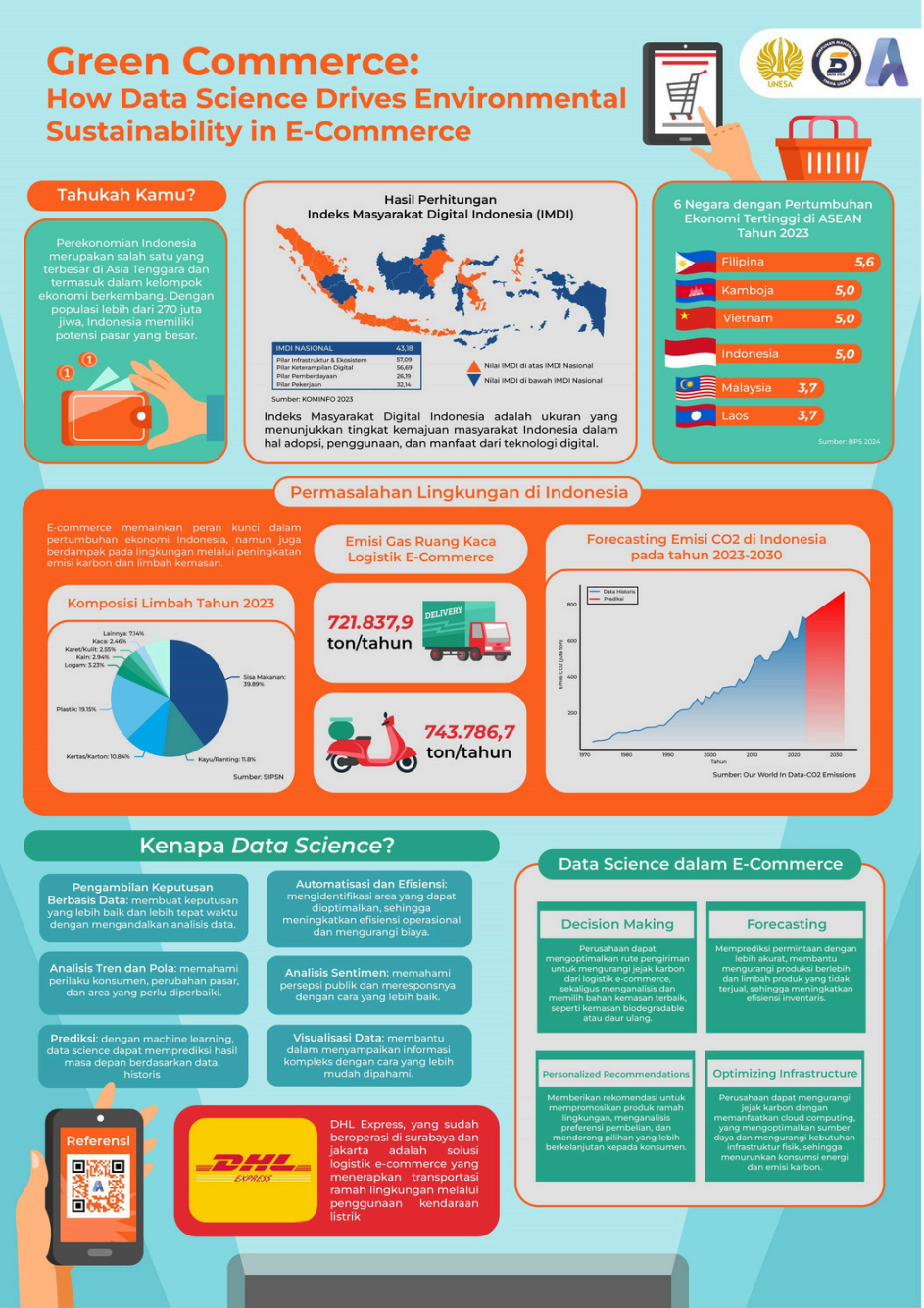
Created a compelling data journalism piece dissecting political hoaxes during the 2024 Indonesian election using clustering, word cloud analysis, and time series charts. Applied advanced data filtering and visualization techniques to expose misinformation patterns, highlight media platforms with high hoax incidence, and communicate verified facts in an accessible narrative format.

Infographic – Smart Living di IKN: Inovasi dalam Kesehatan, Kesejahteraan, dan Keselamatan Publik melalui Data dan Statistik



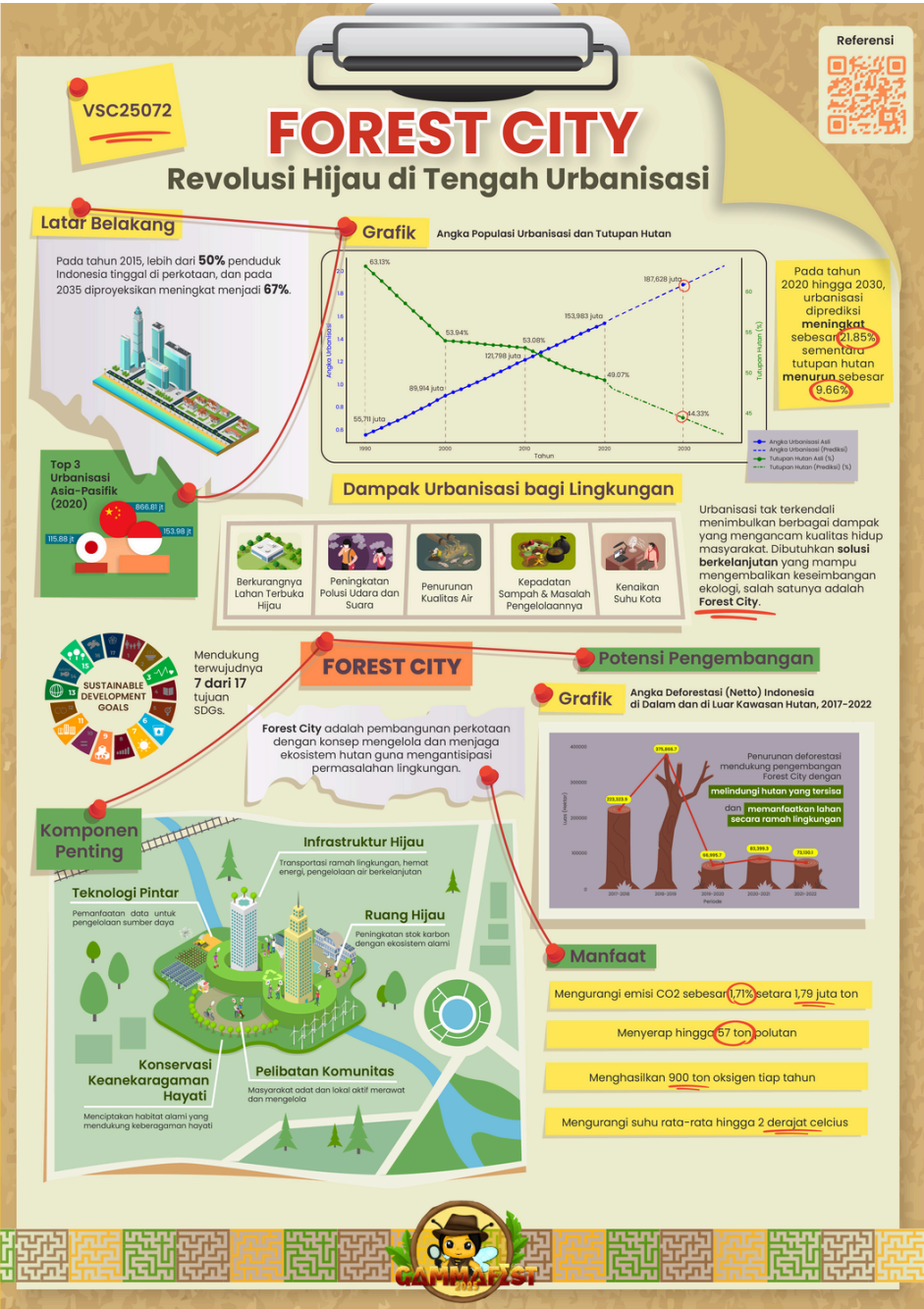
Created a data-driven infographic illustrating the Smart Living concept in Indonesia’s new capital (IKN) using official statistics on digital access, emergency response, and healthcare. Visualized trends in digital society growth, IoT integration, and SDG alignment to show how data and technology improve public safety, health services, and urban planning through clear, engaging visual storytelling.

Infographic – Green Commerce: How Data Science Drives Environmental Sustainability in E-Commerce



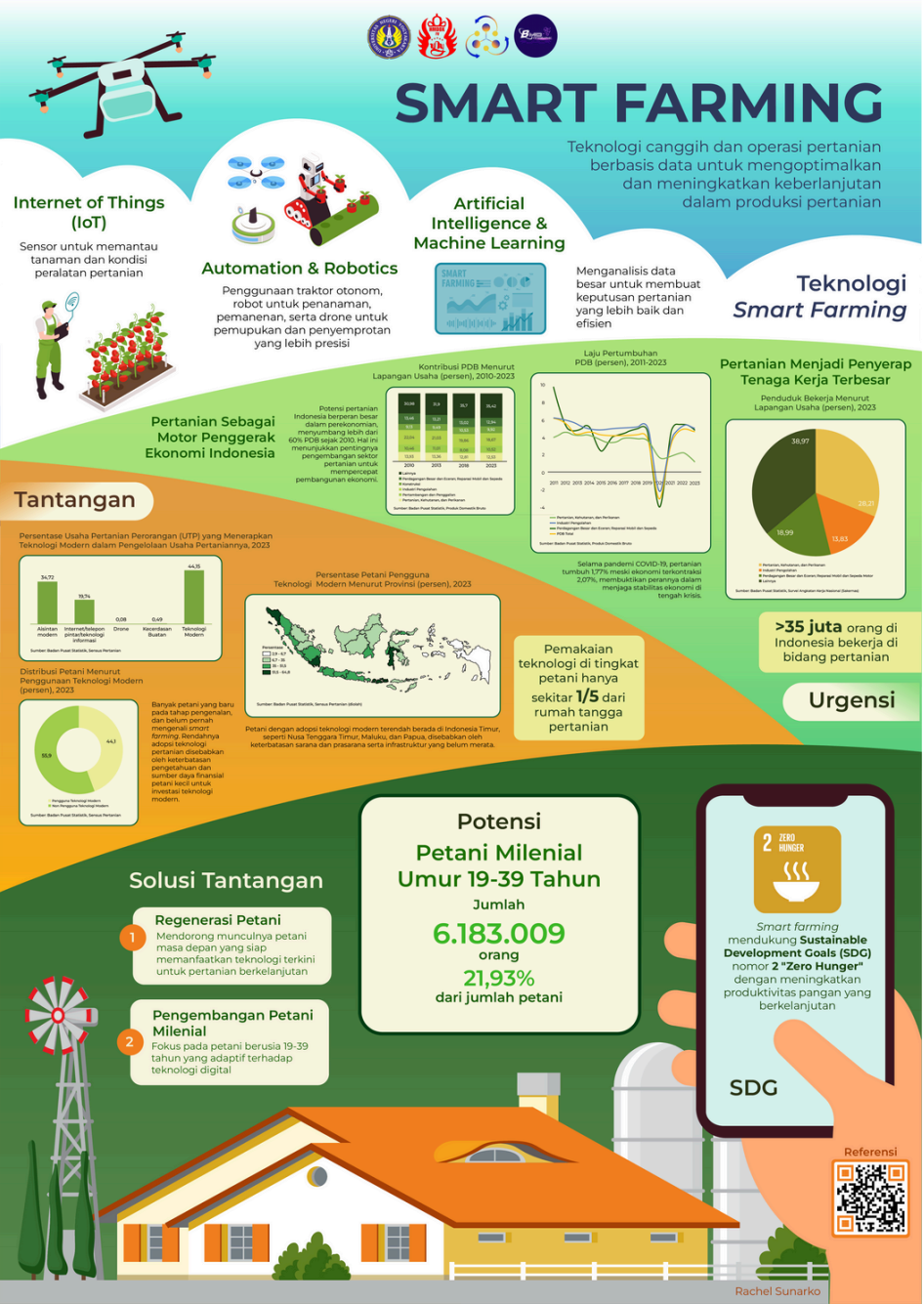
Created a visual storytelling piece on Green Commerce, showcasing how data science can drive environmental sustainability in Indonesia’s e-commerce sector. Leveraged national statistics (IMDI, BPS, SIPPN) to highlight trends in digital adoption, waste composition, and CO2 emissions forecasting. Translated complex data into visual formats like heatmaps, pie charts, and projections to demonstrate the role of data-driven insights in optimizing logistics, reducing emissions, and promoting eco-friendly practices.

Infographic – Forest City: Revolusi Hijau di Tengah Urbanisasi



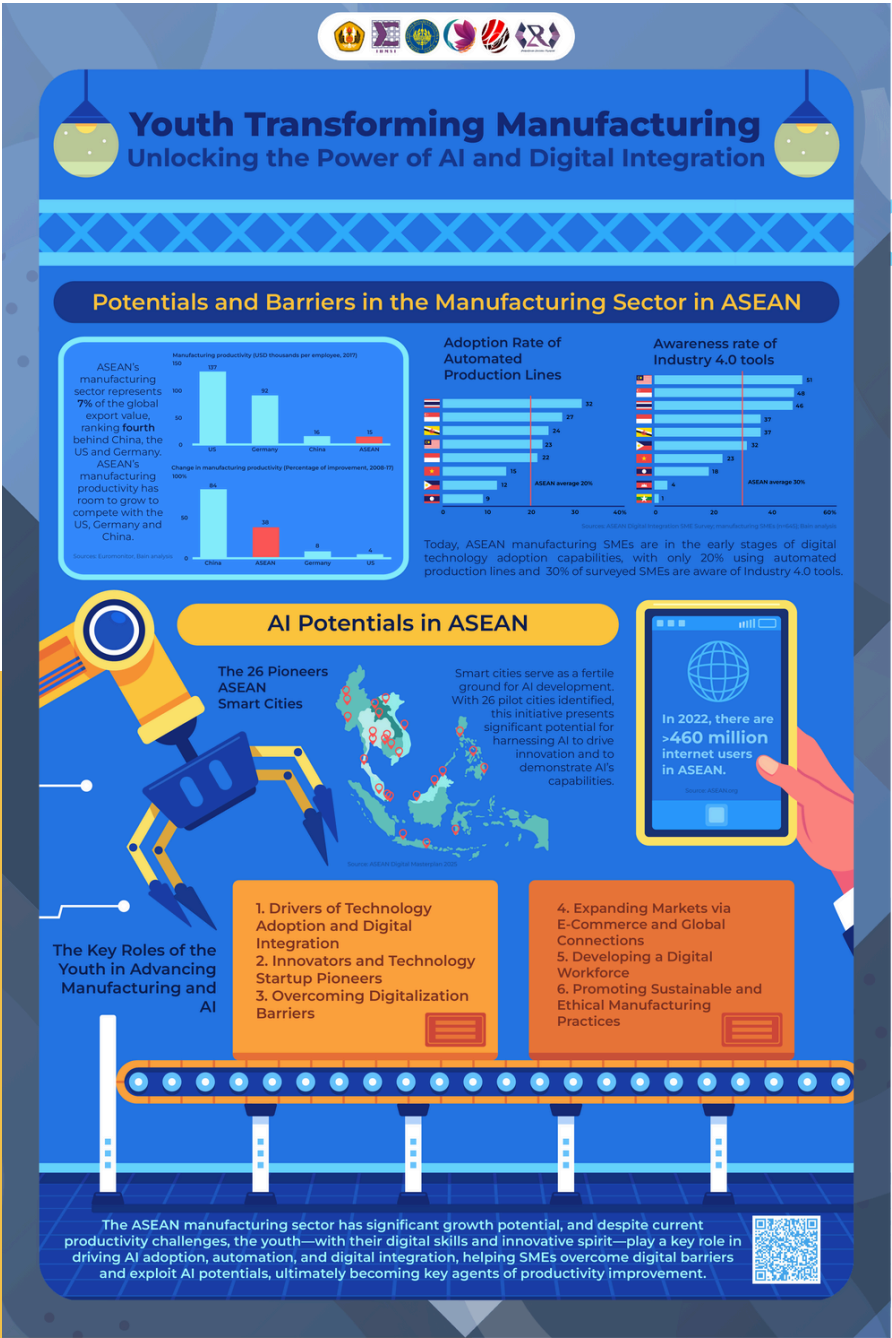
Created a compelling infographic to communicate the urgent need for sustainable urban planning through the "Forest City" concept. Analyzed multi-source datasets on urban population growth, forest cover decline, and deforestation trends to emphasize the ecological impact of rapid urbanization. Translated complex environmental metrics—such as a projected 21.85% rise in urbanization and a 9.66% forest loss—into intuitive visuals, connecting them with actionable urban solutions. Effectively integrated narrative elements and infographics to demonstrate how green infrastructure, community engagement, and biodiversity conservation can support climate goals and Sustainable Development Goals (SDGs).

Infographic – Smart Farming



Created an infographic narrative on Smart Farming by translating agricultural, demographic, and economic data into a compelling visual format. Mapped national trends in modern technology adoption among farmers, identified regional disparities, and emphasized the potential of 6.18 million millennial farmers. Combined geospatial data, time series, and sectoral statistics to highlight the urgency of digital transformation in agriculture and its alignment with SDG 2 ("Zero Hunger").

Infographic – Youth Transforming Manufacturing: Unlocking The Power of AI and Digital Integration



Created a visually engaging infographic analyzing the transformative role of youth in ASEAN’s manufacturing sector through AI and digital integration. Synthesized data on productivity rates, automation adoption, and Industry 4.0 awareness across countries to highlight regional disparities and opportunities. Combined geospatial mapping, comparative bar charts, and strategic insights to narrate how digital-savvy youth can overcome technological barriers, drive innovation, and unlock the sector’s AI potential.

Contact Information

Rachel Sunarko

ADDRESS

Apartemen Puncak Dharmahusada,
Kalijudan, Surabaya

EMAIL

sunarkorachel@gmail.com

PHONE

085701933591
